CCS4711 CRYPTOGRAPHY AND APPLICATIONS

MID TERM TEST

FRIDAY, 24 APRIL 2026

10:30 - 12:00 HOURS

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Any any **TWO** questions from three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

Question 1

A **mobile banking application server** and a **payment gateway server** need to establish a shared secret key before transmitting encrypted transaction data. To do this, the two systems use the **Diffie-Hellman key exchange algorithm**.

The two systems agree on the following public values:

- P=23
- G=5

The **mobile banking server** selects **6** as its private key, while the **payment gateway server** selects **15** as its private key.

As a systems security analyst, compute:

1. The public key generated by the mobile banking server
2. The public key generated by the payment gateway server
3. The shared secret key established between the two systems

Show all your workings.

(25 Marks)

Question 2

A secure messaging system uses the RSA algorithm to protect short text messages sent between systems. For one session, the system is configured with the following values:

- p=17
- q=11
- d= 23
- e=7

The system needs to send the three ASCII characters:

- **A = 65**
- **B = 66**
- **C = 67**

Using the RSA algorithm, answer the following:

1. Calculate the modulus n.
2. Determine the **public key**.
3. Determine the **private key**.
4. Encrypt the ASCII values **65, 66, and 67** using the public key.
5. Decrypt the resulting ciphertext values using the private key.
6. Show that the original ASCII values are recovered after decryption.

(25 Marks)

Question 3

a) ZCAS university wants to issue academic transcripts electronically so that employers can verify they are authentic. Design a solution using digital signatures and explain how the university, graduates, and employers would interact with the system. **(15 marks)**

b) A charity wants donors to trace how their money is used from donation to final expenditure. Propose a blockchain-based system and evaluate whether it would truly improve transparency and trust. **(10 marks)**